

Estimating the Determinants of Market Illiquidity

Learning Objectives:

- What a price impact regression is and how it can be used to measure the various sources of market illiquidity
- How to measure the permanent impact of trades
- How to estimate the level of informed trading in a security

5.1 Introduction

Liquidity varies dramatically from security to security and over time. As Chapters 3 and 4 explain, these variations should reflect changes in the cost of providing liquidity caused by adverse selection, order processing, and risky inventory holding.

The financial crisis of 2008–9 provides a case in point: during the crisis, most stocks became significantly more illiquid. As an illustration, Figure 5.1 plots the average closing bid-ask spread for NYSE stocks in the financial and consumer staples sectors from April 2006 to April 2009. The spreads start rising in all sectors as the first signs of trouble appear in the summer of 2007, and keep increasing as the crisis deepens, especially after the AIG bailout and the Lehman Brothers bankruptcy in the early fall of 2008.

The crisis also generated a huge increase in the volatility of returns by amplifying uncertainty about public policies and companies' prospects. Moreover, it made it harder for liquidity suppliers to get credit to fund their market-making activity. For these reasons, their inventory holding costs increased, which provides a first explanation for the increase in spreads in the sectors of financials and consumer staples.

In addition, the crisis increased asymmetric information for stocks in the financial sector. By their very position, participants from the financial industry were probably better informed about the soundness of the balance sheets of banks and security firms and the likelihood of government bailouts. Thus, the

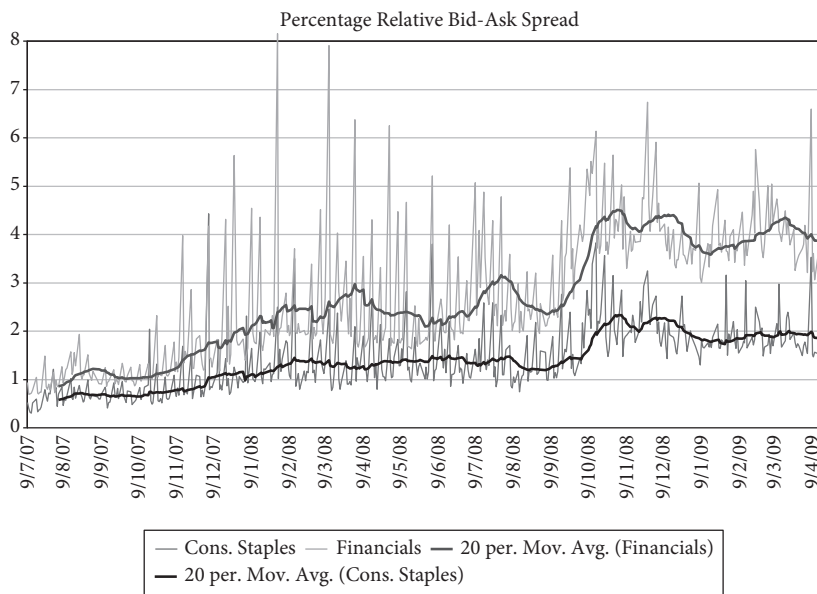


Figure 5.1 Average NYSE closing bid-ask spread, financials, and consumer staples, July 2007–September 2009.

likelihood of informed trading in financials was now greater than before, which may explain why the crisis widened the bid-ask spread on financials more than on consumer staples stocks.

The role of asymmetric information in explaining variations in market liquidity is also illustrated by Figure 5.2, which compares the bid-ask spread for firms that lose analyst coverage (the black line) and firms that do not (the gray line). Financial analysts collect, produce, and disseminate information about the fundamental value of securities. This should level the informational playing field between market participants and thus reduce the opportunities for company insiders. So when analysts' coverage of a company ceases, we expect the bid-ask spread to widen due to adverse selection, and this is precisely what Figure 5.2 shows: when a stock loses its last analyst, its bid-ask spread increases dramatically compared to those of comparable stocks that do not.

The theories set forth in previous chapters, then, have given us a *qualitative* interpretation of cross-sectional and time variations in market liquidity. As we shall see in this chapter, they also serve to *quantify* the contribution of each source of market illiquidity to these variations, using econometric techniques. In Sections 5.2 and 5.3, we explain how one can use time series of transaction prices and signed trade sizes to estimate the contributions of adverse selection

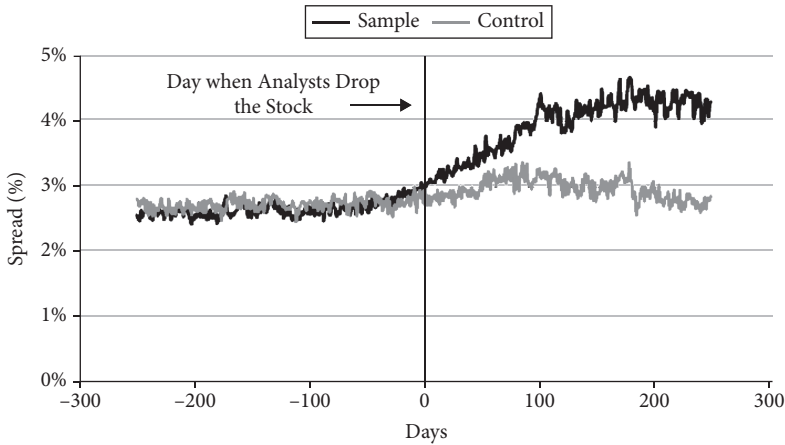


Figure 5.2 Average effective bid-ask spreads for stocks for which analysts coverage is terminated and a control group for which it is continued.

Source: Ellul and Panayides, 2018.

costs, inventory costs, and order processing costs to market illiquidity and price movements. The techniques described exploit the fact that market orders have different effects on the dynamics of securities prices depending on the relative prevalence of adverse selection, order processing, and inventory holding costs (see Chapter 3). Finally, in Section 5.4, we show how one can use order flow data to build a measure of the frequency of informed trading in a security. Armed with this gauge, one can test predictions about the effect of informed trading on liquidity and the sources of informed trading (e.g., whether some trading mechanisms are more prone to informed trading).

5.2 Price Impact Regressions

In this section, we show how to estimate the relative importance of the various costs of providing liquidity by running “price impact regressions,” regressions of changes in prices on contemporaneous and lagged measures of order flow. The exact specification of these regressions depends on assumptions about the sources of market illiquidity and the properties of the order arrival process. In Section 5.2.1, we consider the case in which dealers are risk neutral (so as to ignore inventory holding costs) and market orders are serially uncorrelated. Then, Section 5.2.2 explains how to account for inventory holding costs and serial correlation in specifying price impact regressions.

5.2.1 Without Inventory Costs

Consider again the model of trading developed in Section 3.4: liquidity suppliers are risk neutral and they risk trading with better informed investors. They also bear an order-processing cost for each trade. In Chapter 3's scenario, the price of the t^{th} transaction is $p_t = \mu_t + \gamma d_t$, as in equation (3.33), where (i) μ_t is liquidity suppliers' estimate of the asset payoff, (ii) γ is the order-processing cost per share (or a measure of dealers' rent), and (iii) d_t indicates whether the market order triggering the t^{th} transaction is a buy ($d_t = 1$) or a sell ($d_t = -1$).

Accordingly, consecutive price changes are:

$$\Delta p_t = p_t - p_{t-1} = \mu_t - \mu_{t-1} + \gamma \Delta d_t. \quad (5.1)$$

As orders contain information, they affect liquidity suppliers' estimate of the asset's true value (see equation (3.24) in Chapter 3). Specifically:

$$\mu_t = \mu_{t-1} + \lambda d_t + \varepsilon_t, \quad (5.2)$$

where λ measures the informativeness of the order flow,¹ and the error term ε_t is a white noise error that captures the effects of public information (e.g., macroeconomic and corporate announcements) on market participants' value estimates: hence, $E(\varepsilon_t) = 0, \forall t$, and $E(\varepsilon_t \varepsilon_s) = 0, \forall t \neq s$.

Thus, the transaction price can be written as $p_t = \mu_{t-1} + (\lambda + \gamma)d_t + \varepsilon_t$. Recalling that $p_t = a_t$ if $d_t = 1$ and $p_t = b_t$ if $d_t = -1$, the bid-ask spread at the time of the t^{th} transaction is equal to:

$$a_t - b_t = 2(\gamma + \lambda), \quad (5.3)$$

where λ and γ are respectively the adverse selection cost component and the order processing cost component of the bid-ask spread.

Combining equations (5.1) and (5.2) yields:

$$\Delta p_t = \gamma \Delta d_t + \lambda d_t + \varepsilon_t. \quad (5.4)$$

This equation shows that price movements over a short time interval can arise from three sources: the bid-ask bounce due to order-processing costs, the information content of the order flow, and public news. For example, suppose a buy order arrives at time $t - 1$ and then a sell order at time t . The buy order

¹ The parameter λ corresponds to $s(d_t)$ in equation (3.24), which is time varying, but for simplicity is assumed to be constant, at least over the estimation period.

executes at the best ask price, the sell order executes at the best bid. This “bid-ask bounce” generates a price decrease equal to 2γ , which is captured by the first term ($\gamma\Delta d_t$) in equation (5.4). Moreover, dealers set the bid price for the t^{th} transaction at a discount from their value estimate after the $(t-1)^{\text{th}}$ transaction, reflecting the possibility that the t^{th} order is from an informed investor. This discount also contributes to the change in price between the two transactions, and is captured by the second term (λd_t) in equation (5.4). In this case, it amplifies the price decrease due to the bid-ask bounce by an amount equal to λ . Thus, in this example the total change in price from the $(t-1)^{\text{th}}$ to the t^{th} transaction is $\Delta p_t = -\lambda - 2\gamma < 0$, as implied by equation (5.4). By the same token, in the presence of informed trading, Roll’s measure underestimates the bid-ask spread, being solely based on the bid-ask bounce (see Box 5.1).

Box 5.1 Revisiting Roll’s Measure with Adverse Selection

As Chapter 2 demonstrates, we can use the covariance of consecutive changes in prices to estimate the bid-ask spread. For instance, Roll’s estimator of the bid-ask spread $S_{\text{Roll}} = 2\sqrt{-\text{cov}(\Delta p_t, \Delta p_{t-1})}$ is. As was first noted by Glosten (1987), when the bid-ask spread is in part due to adverse selection costs, this estimator is biased.

To see why, consider again equation (5.4), where only the direction and not the size of trades matters. As d_t and ε_t are independent of each other and over time, equation (5.4) implies that:

$$\text{cov}(\Delta p_t, \Delta p_{t-1}) = -\gamma(\gamma + \lambda).$$

Thus, Roll’s estimate of the bid-ask spread is:

$$\hat{S}_{\text{Roll}} = 2\sqrt{\gamma(\gamma + \lambda)},$$

whereas the true spread is $S = 2(\lambda + \gamma)$. The Roll estimator underestimates it by a factor of $\sqrt{(\lambda + \gamma)/\gamma}$. But if S is observed directly by looking at quoted prices in addition to transaction prices, then Roll’s measure can be used to estimate the fraction of it that is due to order-processing costs, since $\hat{S}_{\text{Roll}}/S = \frac{2\sqrt{\gamma(\lambda + \gamma)}}{2(\lambda + \gamma)} = \sqrt{\frac{\gamma}{\lambda + \gamma}}$. Thus, the order-processing cost component accounts for a fraction $(\hat{S}_{\text{Roll}}/S)^2$ of the spread.

More generally, price reversals are a symptom of market illiquidity. For this reason, $-\text{cov}(\Delta p_t, \Delta p_{t-1})$ (a measure of the magnitude of price reversals)

is sometimes used as a gauge of market illiquidity. However, if illiquidity is mainly due to asymmetric information, this measure of illiquidity will be small even if the spread is large (since in this case $\text{cov}(\Delta p_t, \Delta p_{t-1})$ goes to zero when γ goes to zero). The reason is that the price impact of informed trades is permanent, unlike that of trades due to order-processing costs or inventory holding costs.

Equation (5.4) is the cornerstone of what empiricists call a “price impact regression.” Indeed, with trade and quote data like those presented in Table 3.1, the parameters in this equation can be estimated by running a regression of the trade-to-trade changes in price on the contemporaneous order flow (d_t) and the first difference of this variable (Δd_t). The term “price impact” reflects the fact that the estimates of λ and γ provide a way to evaluate the short-term impact ($\lambda + \gamma$) and the long-term impact (λ) of market orders on transaction prices (see Chapter 3). Hence, this methodology enables to assess the relative importance of the various factors affecting price changes over a short period of time.

In general, researchers have used more complex specifications for price impact regressions than that suggested by equation (5.4). First, the order-processing cost may vary with the size of trades. For instance, if this cost has a fixed component, dealers may be willing to execute large orders at smaller markups or discounts. Alternatively, they may offer better quotes to clients making large trades, as those clients have more bargaining power (see exercise 5 in Chapter 10).² These effects work to create a negative relationship between the effective spread and trade size. To account for these “quantity discounts,” equation (5.1) can be rewritten as follows:

$$\Delta p_t = \mu_t - \mu_{t-1} + \gamma_0 \Delta d_t + \gamma_1 \Delta q_t, \quad (5.5)$$

where γ_1 should be negative in presence of quantity discounts.

Moreover, Chapter 4 shows that not only the direction of orders d_t but also their signed size $q_t = d_t |q_t|$ is informative, since the size of an informed trader’s order will increase in the extent of the deviation between his estimate of the asset’s value and the prior estimate made by liquidity suppliers (see Section 4.2.3). In this case, it is natural to generalize equation (5.2) in this way:

² For instance, institutional investors trade larger volumes than retail investors. As they trade more frequently, they have more bargaining power (see Bernhardt et al. 2005). This effect implies that the bid-ask spread may decrease with trade size, a conclusion at odds with the prediction of asymmetric information models but consistent with empirical observations in bond markets (see Green, Hollifield, and Schürhoff 2007) and some equity markets (e.g., Bernhardt et al. 2005).

$$\mu_t = \mu_{t-1} + \lambda_0 d_t + \lambda_1 q_t + \varepsilon_t. \quad (5.6)$$

The coefficient λ_0 should be zero if very small trades have no informational content. Combining equations (5.5) and (5.6), we obtain a more general version of equation (5.4):

$$\Delta p_t = \lambda_0 d_t + \lambda_1 q_t + \gamma_0 \Delta d_t + \gamma_1 \Delta q_t + \varepsilon_t. \quad (5.7)$$

In estimating such a price impact regression, several econometric problems arise.³ First, data do not always tell whether the trades are triggered by buy or by sell orders (that is, the trade direction d_t , hence the sign of q_t is not always observed). In this case, one must infer market order direction from recorded trades (by using the Lee-Ready algorithm described in Chapter 2), which can induce measurement error in d_t .⁴ Second, real-world quotes must be positioned on a discrete grid, so the observed transaction price necessarily differs from the theoretical price by a rounding error. This is a source of error in the regression's dependent variable, which tends to induce negative autocorrelation in residuals. Last, the error term in this regression is likely to be heteroskedastic if, for instance, public information does not arrive at a constant rate during the day: this biases the estimated standard errors of the coefficients.

There are several ways of coping with these econometric problems. One way is to estimate equation (5.7) and an order direction equation jointly, using a maximum likelihood approach, as described in Glosten and Harris (1988), even though this approach retains the assumption that the error term is homoskedastic. In some cases, data sets are richer, so that order direction is observed directly—as in de Jong, Nijman, and Röell (1996). These authors deal with heteroskedasticity and autocorrelation of residuals using Newey-West estimates of the standard errors of the coefficients. Another method is that adopted by Huang and Stoll (1997) and Madhavan, Richardson, and Roomans (1997), who estimate an equation similar to (5.7) using the generalized method of moments (GMM), which can accommodate conditional heteroskedasticity of unknown form and serial correlation of residuals.

Glosten and Harris (1988) estimate equation (5.7) using a sample of eight hundred transaction-by-transaction observations for NYSE stocks between

³ The specification of the regression is a problem in itself as the omission of a single relevant explanatory variable can bias the estimates. This underscores the need for theoretical models of market microstructure, such as those developed in Chapter 3. They provide a starting point to think about economically meaningful specifications of the relationships between returns and order flow.

⁴ Errors in measurement of the explanatory variables can lead to biased and inconsistent estimates of the coefficients in the classical linear regression model.

December 1, 1981, and January 31, 1983.⁵ They first test the parameter restrictions $\lambda_0 = \gamma_1 = 0$, namely, that small orders carry no information and that order-processing costs are unrelated to trade size. Upon finding that this restriction is not rejected on a subsample, they estimate the following restricted specification of equation (5.7) for each of 250 stocks:

$$\Delta p_t = \lambda_1 q_t + \gamma_0 \Delta d_t + \varepsilon_t. \quad (5.8)$$

They find that λ_1 is significantly positive for 170 stocks and γ_0 is significantly positive for 210 stocks. On average, for the stocks in their sample, they find that for a one-thousand share lot, $\lambda_1 = 0.0102$ and $\gamma_0 = 0.0465$.⁶ To interpret these figures, consider a typical stock and suppose that a buy market order for one thousand shares arrives. According to the model, on average, this order executes at a markup (relative to the midquote) equal to: $\gamma_0 + \lambda_1 = 5.67$ cents per share. Thus, the total implicit trading cost on this order is $1,000 \times 0.0567 = \$56.70$. However, after such a trade, the price increases permanently by an average amount equal to 1.02 cents (λ_1). So the liquidity suppliers' expected profit on the transaction (the "realized spread") is only 4.65 cents per share. This example shows how one can use price impact regressions to obtain estimates of effective and realized spreads (see Chapter 2 for their definitions).

Using different data, de Jong, Nijman, and Röell (1996) find that on average the adverse selection component of the spread is weakly increasing in order size, as in Glosten and Harris (1988), but unlike the latter they find that the order-processing cost component is strongly decreasing in trade size (i.e., $\gamma_1 < 0$).

5.2.2 With Inventory Costs

If liquidity suppliers require compensation for inventory risk, as explained in Section 3.5, the price impact regressions (5.4) and (5.7) are misspecified. They will accordingly produce biased estimates of order-processing and adverse selection costs. Indeed, as shown graphically in Figure 3.11 of Chapter 3, part of the change in liquidity suppliers' quotes after a trade reflects the change in inventory. This will dissipate over time as their aggregate inventory reverts to its long-term average. But in the medium term, this impact persists and moves quotes in the same direction as the trade. So the coefficient of the contemporaneous order

⁵ Our notation differs from that used by Glosten and Harris (1988). They denote by z_0 and z_1 the coefficients that we denote by λ_0 and λ_1 , and by c_0 and c_1 the coefficients that we denote by γ_0 and γ_1 .

⁶ See table 2, panel B on p. 136 of their study.

flow in equation (5.4) will pick up both the impact of asymmetric information and that of inventory risk, and will therefore overestimate the informativeness of order flow.

To see this point, let us expand the model to include not only asymmetric information but also inventory holding costs. For simplicity, we assume that the informational content of the order flow is entirely contained in the signed trade size q_t (so that $\lambda_0 = 0$ in equation (5.6)):

$$\mu_t = \mu_{t-1} + \lambda q_t + \varepsilon_t. \quad (5.9)$$

This means that just before observing the order flow q_t , dealers estimate the security's value as $\mu_{t-1} + \varepsilon_t$, assuming that they expect the order flow to be zero on average. Factoring in inventory holding costs, their valuation before executing the order at time t is:

$$m_t = \mu_{t-1} - \beta z_t + \varepsilon_t, \quad (5.10)$$

where z_t is dealers' aggregate inventory just before the t^{th} transaction and β reflects inventory holding costs, that is, it equals the risk premium parameter $\rho\sigma_\varepsilon^2$ in equation (3.45) of Chapter 3.

Taking the first differences of the midquote in equation (5.10), we obtain:

$$\Delta m_t = \Delta\mu_{t-1} - \beta\Delta z_t + \Delta\varepsilon_t = \lambda q_{t-1} - \beta\Delta z_t + \varepsilon_t, \quad (5.11)$$

where in the second step we have substituted $\Delta\mu_{t-1} = \lambda q_{t-1} + \varepsilon_{t-1}$ from equation (5.9). Using the market-clearing condition $\Delta z_t = -q_{t-1}$ (in equilibrium, the change in inventories mirrors the order flow), we can rewrite equation (5.11) as:

$$\Delta m_t = (\lambda + \beta)q_{t-1} + \varepsilon_t. \quad (5.12)$$

Now the transaction price at time t is the midquote plus or minus half the bid-ask spread, depending on whether the incoming order is a buy or a sell, that is,

$$p_t = m_t + (\lambda + \beta)q_t + \gamma d_t, \quad (5.13)$$

where we assume that the order-processing cost is γ per share. Taking the first difference of this equation and using equation (5.12), the change in consecutive prices can be written as:

$$\Delta p_t = (\lambda + \beta)q_t + \gamma\Delta d_t + \varepsilon_t. \quad (5.14)$$

Thus even if the regression (5.8) estimated by Glosten and Harris (1988) is correctly specified, the estimate of the coefficient of q_t will overestimate the informativeness, λ , of the order flow in the presence of inventory effects. The problem is that this specification is not rich enough to distinguish the adverse selection and the inventory cost components, because in the short run they have the same effect on prices.

To overcome this identification problem, one can enrich the model to account for the fact that market orders are serially correlated, as suggested by Huang and Stoll (1997). As explained below, taking this into account allows a decomposition of the bid-ask spread in its three components: adverse selection, order-processing costs and inventory-holdings costs. Serial correlation in the flow of market orders may arise for at least three reasons. First, informed investors often react to the same signal, which generates a flurry of orders on the same side of the market. This generates positive serial correlation in order flow (see exercise 1). Second, an investor who wants to make a large trade may elect to trickle it into the market as a series of smaller orders to lessen its price impact. This also tends to generate positive serial correlation in the order flow. Third, as Chapter 3 explains, dealers react to an unbalanced order flow by changing their quotes to induce more orders from the opposite side of the market and so rebalance their inventories. This induces negative autocorrelation in the order flow.

Regardless of its sign, the autocorrelation implies that there is a predictable component of the order flow. Let $\eta_t \equiv q_t - E(q_t | \Omega_{t-1})$ be the unexpected component (i.e., the “innovation”) in the t^{th} transaction (Ω_{t-1} being all the relevant information available before the t^{th} transaction relevant for predicting its size). As Hasbrouck (1988) observes, only this unexpected component contains new information and therefore affects dealers’ value estimate. Thus equation (5.9) becomes (see exercise 1 for a derivation in the framework analyzed in Chapter 3):

$$\mu_t = \mu_{t-1} + \lambda [q_t - E(q_t | \Omega_{t-1})] + \varepsilon_t. \quad (5.15)$$

Recalling that the transaction price is the midquote plus or minus half the bid-ask spread, we can rewrite the transaction price at time t as follows:

$$p_t = m_t + \lambda [q_t - E(q_t | \Omega_{t-1})] + \beta q_t + \gamma d_t, \quad (5.16)$$

where m_t is given by equation (5.10). The adverse selection cost component of the spread (that is, $\lambda [q_t - E(q_t | \Omega_{t-1})]$) depends only on the trade innovation for the reasons that we just explained. But the inventory cost component (βq_t) is determined by the actual trade size, since the change in dealers’ inventory (and thereby their inventory risk exposure) depends on the entire (not just the unexpected) trade size. This provides a way to measure separately the

contribution of asymmetric information (λ) and inventory risk (β) to market illiquidity.

As an example, let us assume, as in Huang and Stoll (1997), that market orders are generated by a first-order autoregressive process:

$$q_t = \phi q_{t-1} + \eta_t, \quad (5.17)$$

where the innovation η_t in the order flow is assumed to be independent of current information Ω_{t-1} . Hence, the expected value of the order flow is $E(q_t|\Omega_{t-1}) = \phi q_{t-1}$. In this case the order flow innovation $q_t - E(q_t|\Omega_{t-1})$ is $q_t - \phi q_{t-1}$. Inserting this innovation in expression (5.16) and taking first differences yields the change in transaction prices:

$$\Delta p_t = \Delta m_t + \lambda [\Delta q_t - \phi \Delta q_{t-1}] + \beta \Delta q_t + \gamma \Delta d_t. \quad (5.18)$$

As in the first step of equation (5.11), the change in the midquote is $\Delta m_t = \Delta \mu_{t-1} - \beta \Delta z_t + \Delta \varepsilon_t$. Again using the market-clearing condition $\Delta z_t = -q_{t-1}$ and the fact that the change in dealers' value estimate is:

$$\Delta \mu_t = \lambda (q_t - \phi q_{t-1}) + \varepsilon_t, \quad (5.19)$$

we obtain:

$$\Delta m_t = (\lambda + \beta) q_{t-1} - \lambda \phi q_{t-2} + \varepsilon_t. \quad (5.20)$$

Substituting this expression for Δm_t in equation (5.18), we finally obtain the following specification for the price impact regression:

$$\Delta p_t = (\lambda + \beta) q_t - \lambda \phi q_{t-1} + \gamma \Delta d_t + \varepsilon_t. \quad (5.21)$$

When estimated jointly with the autoregressive process for the order flow (5.17), this specification of the price impact regression allows us to identify the three determinants of the bid-ask spread: λ, β, γ , and the coefficient of the order flow process, ϕ . Hence, this approach is called a *three-way decomposition* of the bid-ask spread. Identification comes from the additional assumptions that dealers' estimate of the fundamental value is affected only by surprises in order flow and that inventory holding costs are affected by the actual order flow.

Madhavan, Richardson, and Roomans (1997) estimate a special case of equation (5.21), assuming that $q_t = d_t$ (i.e., all trades are posited as being for one share) and $\beta = 0$ (no inventory effect). So, while their approach distinguishes only between the adverse selection and order-processing cost components of the bid-ask spread, as already done by Glosten and Harris (1988), it takes

into account that only the unforeseen component of the order flow conveys information. In their case, equation (5.21) becomes:

$$\Delta p_t = (\lambda + \gamma) d_t - (\lambda \phi + \gamma) d_{t-1} + \varepsilon_t. \quad (5.22)$$

Using GMM they estimate this equation jointly with (5.17) (to estimate ϕ) for various intervals during the trading day for NYSE stocks. They show that the adverse selection cost component is relatively high at the opening (about 4 cents in the first half hour of trading; see their table 2) and then declines (to about 2.8 cents in the last half hour). In contrast, the order-processing cost component is lower at the opening (about 3.4 cents) than at the end of the day (about 4.6 cents). This pattern may reflect the fact that dealers' bargaining power increases towards the end of the trading day, since liquidity demanders are more and more impatient to trade. Hence the daily evolution of the adverse selection cost and order-processing cost implies a U-shaped curve of bid-ask spreads in equity markets: high at the beginning and the end of the trading day, low in the middle.

When data on dealers' inventories are available, there is another way to estimate the various components of the bid-ask spread and to relate price changes to order flow. To see this, add and subtract βq_{t-1} from the right hand side of equation (5.21) to obtain:

$$\begin{aligned} \Delta p_t &= (\lambda + \beta) q_t - \beta q_{t-1} + \beta q_{t-1} - \lambda \phi q_{t-1} + \gamma \Delta d_t + \varepsilon_t \\ &= \lambda q_t + \beta (q_t - q_{t-1}) - (\lambda \phi - \beta) q_{t-1} + \gamma \Delta d_t + \varepsilon_t \\ &= \theta_0 q_t + \theta_1 (q_t - q_{t-1}) + \theta_2 (z_t - z_{t-1}) + \gamma \Delta d_t + \varepsilon_t, \end{aligned} \quad (5.23)$$

with $\theta_0 = \lambda$, $\theta_1 = \beta$, and $\theta_2 = \lambda \phi - \beta$, after using the market clearing condition $z_t - z_{t-1} = -q_{t-1}$ in the last step of (5.23). Thus, with data on dealers' inventories, we can evaluate the strength of inventory effects by estimating a regression of price changes on inventories and order flow. In practice, the coefficient of $\Delta q = q_t - q_{t-1}$ may also reflect the effect of order-processing costs if these have a fixed component, as explained in the previous section. Thus, the coefficient of the price impact regression on the change in inventories, $z_t - z_{t-1}$, is likely to be more informative about the effect of inventory risk on prices. It should be negative if $\beta > \lambda \phi$ (i.e., if inventory effects are strong enough).

Lyons (1995) tests a specification similar to equation (5.23) with data on trades of one dealer in the Deutsche Mark/Dollar market in 1992 (over one week, August 3–August 7, 1992). For each transaction in which the dealer acts as a liquidity supplier, Lyons observes its size and the dealer's inventory. He finds

very strong inventory effects. One possible reason is that in foreign exchange markets, inventory risk is large, as trades are typically larger there than in equity markets. Moreover, dealers tend to avoid carrying large inventories overnight, so they must adjust their quotes more substantially after trades in order to get faster mean reversion in their positions. Exercise 4 invites the reader to use the data in Lyons (1995) to estimate various specifications of the price impact regressions described in this section.

5.3 Measuring the Permanent Impact of Trades

The previous section models order arrival as an AR(1) process. That is, traders' expectations regarding the next trade depend only on the last trade. Accordingly, the order flow process is assumed to have a very "short memory." This assumption is problematic: with inventory effects, an order's impact is likely to ripple far in the future, suggesting that we should consider autoregressive process with lags greater than one. Consider, say, the arrival of a large sell order at time t . This order leads to a decrease in price, due in part to the increase in liquidity suppliers' risk exposure rather than a true decline in the asset value. Thus, the new order at time $t + 1$ is more likely to be a buy than a sell, which corresponds to the case where $\delta < 0$ in the model of Section 3.5.2. But unless dealers' inventories revert very quickly to their long-run level, prices—even after the next trade—will continue to look relatively low to investors. Thus, at time $t + 2$, a buy order is again more likely than a sell. In fact, this will continue to be the case until dealers' aggregate inventory reverts to its long-run level. This suggests that the trade size at time t should affect expectations about trades at time $t + 1$ but also at $t + 2, t + 3$, and so on. Moreover, as dealers' inventories revert to their long-run level, a reversal in prices will temper the initial negative impact of the sell order. Hence, in the presence of inventory effects, the permanent impact of a trade is smaller than its immediate impact, and the short-run impact takes time to dissipate (see Figure 3.11).

To account for these observations, Hasbrouck (1991) proposes to measure the permanent impact of trades (a measure of their informativeness) by jointly modelling the dynamics of price changes and orders by a vector autoregressive model (VAR). More specifically, changes in midquotes and trades are assumed to behave according to the following processes:

$$\begin{aligned} m_t - m_{t-1} &= \sum_{j=1}^{\infty} a_j \Delta m_{t-j} + \sum_{j=1}^{\infty} b_j q_{t-j} + \varepsilon_t, \\ q_t &= \sum_{j=1}^{\infty} c_j \Delta m_{t-j} + \sum_{j=1}^{\infty} h_j q_{t-j} + \eta_t, \end{aligned} \quad (5.24)$$

with the assumptions that the innovations ε_t and η_t have zero means and are jointly and serially uncorrelated.⁷ If one sets $b_1 = \lambda + \beta$, $b_2 = -\lambda\phi$, $h_1 = \phi$, and all other coefficients to zero, this specification encompasses as a special case the midquote changes given by equation (5.20).

Besides this particular specification, the VAR approach can encompass richer interactions between returns and order flow: per se, this approach does not place any restrictions on these interactions. Typically, these restrictions will follow from an economic model. For instance, in the presence of inventory effects, one would expect the coefficients h_j to be negative, since inventory and quote management induce mean reversion of the order flow. If, instead, investors tend to execute large orders piecemeal, then one would expect a positive autocorrelation in the order flow and positive values for some of the coefficients h_j . Moreover, the order flow could respond to past price movements: for instance, investors may tend to place buy market orders in response to low quotes and sell market orders in response to high quotes. Such contrarian strategies should result in negative coefficients c_j .

As an illustration, consider the following example inspired by Hasbrouck (1991). Suppose that trades and changes in midquotes for a stock behave according to the following model:

$$\begin{aligned} m_t - m_{t-1} &= 1.998q_{t-1} + 0.096q_{t-2} + 0.053q_{t-3} + 0.025q_{t-4} + \varepsilon_t, \\ q_t &= -0.486q_{t-1} - 0.269q_{t-2} - 0.124q_{t-3} + \eta_t. \end{aligned} \quad (5.25)$$

Using this specification, we can compute the dynamics of midquote returns and trades after a purchase of size 1 ($q_0 = 1$) when the innovations in the system are set at their average value, namely, zero. Table 5.1 below shows the dynamics of the midquotes and trades from the first transaction to the tenth transaction. For the purpose of this discussion, we normalize the initial estimate of the asset value to $m_0 = 100$.

On receiving the buy order at time $t = 0$, the dealers adjust his midquote at time $t = 1$ up by 1.998, from 100 to 101.998. Models of trading with adverse selection and inventory holding costs typically attribute this increase to the informational content of the buy order *and* to the decrease in dealers' aggregate inventory. In fact, in the aggregate, dealers have sold one unit of the security and now have a short position, so they mark up the price at which they are willing to sell another unit. Hence, the informational content of the trade can be anywhere from 0 to 1.998 depending on the relative magnitude of adverse selection costs and inventory costs.

⁷ Our notation and timing conventions are different from Hasbrouck (1991).

Table 5.1 Dynamic Responses of Prices and Trades in the Illustrative Model of Hasbrouck (1991)

t	m_t	$m_t - m_{t-1}$	q_t	$\alpha_t(1) = \sum_{\tau=0}^t \Delta m_\tau$
0	100	—	1.000	—
1	101.998	1.998	-0.486	1.998
2	101.124	-0.875	-0.033	1.123
3	101.064	-0.060	0.023	1.064
4	101.105	0.041	0.058	1.105
5	101.210	0.105	-0.030	1.210
...	...	...	...	...
10	101.1566	-6×10^{-4}	-3×10^{-4}	1.156

Source: Joel Hasbrouck (1991), "Measuring the information content of stock trades," *Journal of Finance* 46, 179–207.

Hasbrouck (1991) proposes to identify the information content of the trade with its long-run average price effect (see exercise 2 for an application). The idea is simple: as Chapter 3 explains, a change in the midquote due to inventory holding costs will dissipate over time as dealers reduce their exposure to the desired level by unwinding positions. Hence, the change in price that persists after a trade must correspond to the value estimation of the asset induced by the trade. Accordingly, the average cumulative price change over T periods after a trade of size q_0 (denoted $\alpha_T(q_0)$) can be used as a proxy for the informational content of a trade of this size. In the analysis of VAR models, $\alpha_T(q_0)$ is called an *impulse response function*, in this case to an innovation in order flow. Impulse response functions are routinely computed by standard software for statistical analysis.

As an application of this idea, let us look again at Table 5.1. The cumulative price change from the first to the tenth transaction, $\alpha_{10}(1)$, is 1.156.⁸ This cumulative change in price is less than the initial change ($\alpha_0(1)$), and it can be used as a measure for the informational content of the trade at time $t = 0$. Of course, one can look at an even longer horizon to get a more accurate estimate of the permanent impact of trade (its true informativeness). The choice of horizon should depend on how fast the price impact of a trade that is not due to its information content dissipates.⁹

⁸ To see how $\alpha_t(1)$ is computed, consider the calculation of $\alpha_2(1)$. First we compute q_1 using the equation for the dynamics of q_t : we obtain $q_1 = -0.486 \times q_0 = -0.486$. Substituting this in the equation for the dynamics of Δm_t , we obtain $\Delta m_2 = 1.998 \times q_1 + 0.096 \times q_0 = -1.998 \times 0.486 + 0.096 = -0.875$. Hence, $\alpha_2(1) = \Delta m_1 + \Delta m_2 = 1.998 - 0.875 = 1.123$.

⁹ The econometric model in equation (5.24) assumes that only market orders convey information. However, limit orders may also do so. By extending the methodology of Hasbrouck (1991) to a model

Hasbrouck (1991) implements this methodology using a continuous sequence of trade and quote data from NYSE, AMEX, and consolidated regional U.S. exchanges, over the sixty-two trading days in the first quarter of 1989. He estimates a specification similar to the previous system of equations, truncated at five lags for each variable. The impact of trades on quote revisions is generally positive, and convergence of α_T to its limit value is quite rapid (most being completed within five steps) but not instantaneous. Like other researchers (e.g., Huang and Stoll 1997), Hasbrouck finds that trades are positively autocorrelated (the h_j 's are positive) rather than negatively as expected in the presence of inventory effects. But there is a strong negative correlation with midquote changes (the c_j 's are negative), which is more consistent with the inventory control hypothesis.

The measure of private information features interesting cross-sectional relationships with company size. After scaling the persistent price impact α_T by the corresponding average share price, Hasbrouck finds that the overall price impact of a trade innovation is greater for stocks with lower rather than higher market capitalization, implying that information asymmetry is more severe for the former. This may reflect the lesser availability of public information about smaller firms, which generally have little or no analyst coverage. So this finding is consistent with the evidence of Ellul and Panayides (2011), quoted at the beginning of this chapter, that stocks that lose analyst coverage have lower liquidity.

5.4 Probability of Informed Trading (PIN)

A completely different approach to evaluating the importance of adverse selection in trading is found in the works of David Easley, Maureen O'Hara, and various coauthors. The method that they propose aims at estimating the probability of informed trading (PIN). Their idea is to infer this probability from the volume and the imbalance of buy and sell orders throughout the trading day: for example, if buy market orders prevail, it seems likely that good news has arrived at the start of trading day, prompting informed buying.

Here we present their approach. The starting point is a specification of the order arrival process over a specific period of time which, in most applications, is put at one trading day. Figure 5.3 gives a graphical summary.

On each trading day, there is a probability α that an information event occurs. An information event is a change in the value of the security, which can be either

that includes both market and limit orders, Brogaard et al. (2019) show that limit orders also have permanent impact and therefore contribute to price discovery.

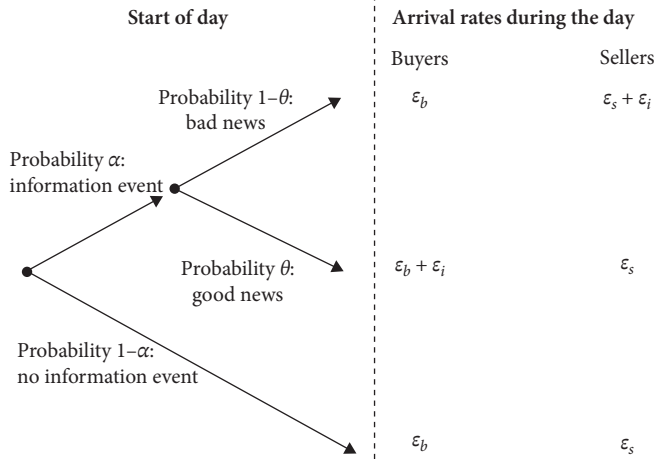


Figure 5.3 Information and order arrival process in the PIN model.

positive with probability θ or negative with probability $1 - \theta$. With probability $1 - \alpha$, there is no information event. A day without an information event is defined as a day without a change in the value of the security.

Dealers do not know whether or not an information event has occurred. If it does, some investors are informed. They know whether the information event is positive (the value of the security rises) or negative. As in Glosten and Milgrom (1985), informed investors will buy on days with good news and sell on those with bad. By definition, there is no informed trading on days without an information event.

More specifically, on days with an information event, orders from informed traders arrive according to a Poisson process with intensity ϵ_i per day. This means that the likelihood of observing k orders from informed traders in a day with an information event is:

$$\frac{\epsilon_i^k e^{-\epsilon_i}}{k!},$$

where $e = 2.71828$ and $k!$ is the factorial of k . Thus, the daily average number of orders from informed traders is ϵ_i . And, whether or not an information event has occurred, buy and sell orders from uninformed investors arrive according to Poisson processes with intensities ϵ_b and ϵ_s , respectively. The order processes for uninformed sellers, uninformed buyers, and informed investors are independent.

The order arrival process differs from the model in Glosten and Milgrom (1985) studied in Chapter 3 in two ways. First, time is continuous and traders

arrive at stochastic points in time (in Glosten and Milgrom (1985), time is discrete). Second, there is uncertainty whether an information event has occurred.¹⁰

As orders from informed and uninformed traders are generated by independent Poisson processes, the likelihood of any given trade being initiated by an informed trader can be shown to be:

$$\text{PIN} = \frac{\alpha \epsilon_i}{\epsilon_b + \epsilon_s + \alpha \epsilon_i}, \quad (5.26)$$

where PIN stands for “probability of informed trading.” As one would expect, PIN is the ratio between the rate of arrival of informed traders and the total rate of order arrival. In the limiting case in which uninformed investors do not trade ($\epsilon_b = \epsilon_s = 0$), then $\text{PIN}=1$. This variable plays the same role as the probability π in Chapter 3.

Using this observation, quotes can be computed as in Chapter 3 (see exercise 7). For instance, consider the bid-ask spread for the opening trade and assume that $\theta = \frac{1}{2}$ and $\epsilon_b = \epsilon_s$, so that uninformed investors are equally likely to buy or sell. In this case, using equation (3.12) in Chapter 3 and replacing π with PIN, we obtain:

$$a_1 - b_1 = 2 \times \text{PIN} \cdot (v_H - v_L), \quad (5.27)$$

where v_H is the realization of the asset’s value on days with good news and v_L on days with bad news.

Thus market illiquidity is positively related with PIN, which measures dealers’ exposure to informed traders. We now explain how to estimate PIN with order flow data. Suppose there are B_n buy and S_n sell market orders on day n . As the order processes of informed and uninformed investors are independent, the likelihood of these realizations conditional on day n being a bad-news day is:

$$\frac{(\epsilon_b)^{B_n} e^{-\epsilon_b}}{B_n!} \cdot \frac{(\epsilon_i + \epsilon_s)^{S_n} e^{-(\epsilon_i + \epsilon_s)}}{S_n!}. \quad (5.28)$$

Conditional on a good-news day, the likelihood of observing B_n buy orders and S_n sell orders is:

$$\frac{(\epsilon_s)^{S_n} e^{-\epsilon_s}}{S_n!} \cdot \frac{(\epsilon_i + \epsilon_b)^{B_n} e^{-(\epsilon_i + \epsilon_b)}}{B_n!}. \quad (5.29)$$

¹⁰ This uncertainty implies that the end value of the security has three possible realizations instead of two as in the model of Chapter 3.

Finally, on a no-news day, the likelihood of observing B_n buy orders and S_n sell orders is:

$$\frac{(\epsilon_s)^{S_n} e^{-\epsilon_s}}{S_n!} \cdot \frac{(\epsilon_b)^{B_n} e^{-\epsilon_b}}{B_n!}. \quad (5.30)$$

Now, recall that the probabilities of the three possible configurations of information are:

$$\text{information} = \begin{cases} \text{no news,} & \text{with probability } 1 - \alpha, \\ \text{bad news,} & \text{with probability } \alpha(1 - \theta), \\ \text{good news,} & \text{with probability } \alpha\theta. \end{cases}$$

Hence, using equations (5.28), (5.29), and (5.30), the unconditional probability of B_n buy orders and S_n sell orders on day n is:

$$\begin{aligned} \Pr(B_n, S_n) = & (1 - \alpha) \frac{(\epsilon_s)^{S_n} e^{-\epsilon_s}}{S_n!} \cdot \frac{(\epsilon_b)^{B_n} e^{-\epsilon_b}}{B_n!} \\ & + \alpha(1 - \theta) \frac{(\epsilon_b)^{B_n} e^{-\epsilon_b}}{B_n!} \cdot \frac{(\epsilon_i + \epsilon_s)^{S_n} e^{-(\epsilon_i + \epsilon_s)}}{S_n!} \\ & + \alpha\theta \frac{(\epsilon_b + \epsilon_i)^{B_n} e^{-(\epsilon_b + \epsilon_i)}}{B_n!} \cdot \frac{(\epsilon_s)^{S_n} e^{-\epsilon_s}}{S_n!}. \end{aligned}$$

Assuming that the occurrence and direction of information events are independent across days, the likelihood of observing a series of daily number of orders B_n and S_n over N days is therefore:

$$\Pr((B_1, S_1), \dots, (B_N, S_N)) = \prod_{n=1}^N \Pr(B_n, S_n). \quad (5.31)$$

This probability is a function of the five structural parameters: $\alpha, \theta, \epsilon_i, \epsilon_b$, and ϵ_s . Thus we can estimate these parameters, and the PIN, by using the maximum likelihood approach.¹¹

Easley, Hvidkjaer, and O'Hara (2002) estimate the PIN of all NYSE common stocks from 1983 to 1998 and obtain a median PIN of about 19%. Their estimate does not vary greatly across stocks (Figure 5.4) and is negatively correlated

¹¹ In order to estimate the PIN, one must know the daily number of buy and sell market orders over the relevant time period. As explained above, some data sets do not identify the direction of market orders, and in these cases, the number of buy and sell orders must be inferred from other variables. This leads to measurement errors that can bias the PIN measure (see Boehmer, Grammig, and Theissen 2006).

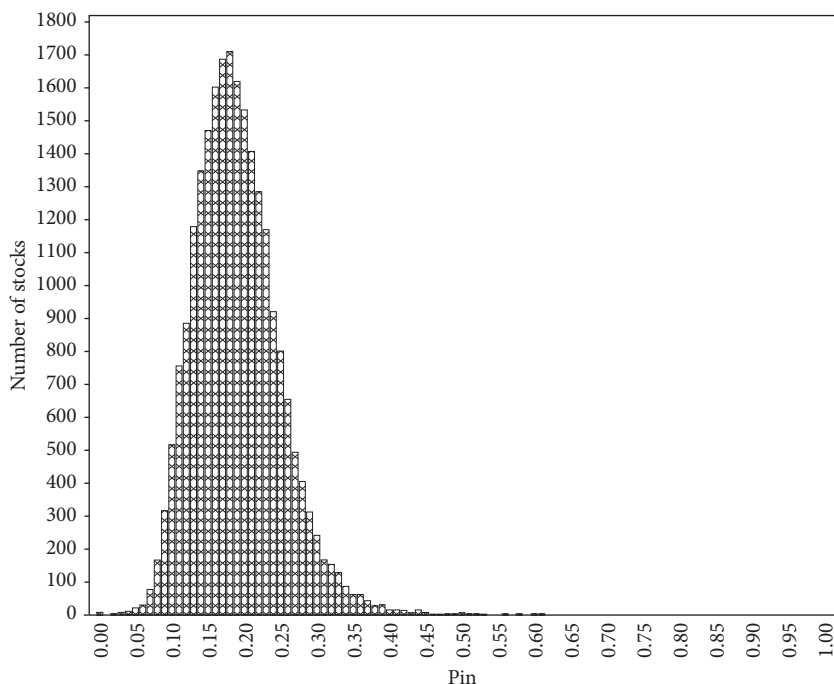


Figure 5.4 Distribution of PIN across NYSE stocks.

Source: panel C of fig. 4 in Easley et al. 2002, p. 2207.

with firm size (with a correlation coefficient of about -0.58): larger firms are characterized by relatively less informed trading, as in Hasbrouck (1991) and many other studies. Meanwhile, the PIN is positively correlated in the cross-section with volatility (0.239) and the bid-ask spread (0.353). The correlation with volatility may reflect the fact that information events are more likely for volatile stocks (α is higher), which also present greater profit opportunities for informed traders (so that ϵ_i is higher for more volatile stocks).

Easley, Hvidkjaer, and O'Hara (2002) also find that the PIN is very stable over time: Figure 5.5 shows the fifth, twenty-fifth, fiftieth, seventy-fifth, and ninety-fifth percentiles of the PIN in each year covered by their study. The PIN does not appear to be correlated with trading volume, perhaps because informed investors adjust their strategies to the trading intensity of noise traders so as to maintain the PIN invariant to volume. Last, as explained in Chapter 10, Easley, Hvidkjaer, and O'Hara (2002) shows that stock returns are positively related to the PIN measure.

This methodology has been applied to many different questions. For instance, Grammig, Schiereck, and Theissen (2001) use it to test the hypothesis that there is less informed trading in non-anonymous trading venues. Intuitively, non-anonymous trading can mitigate informational asymmetries by helping

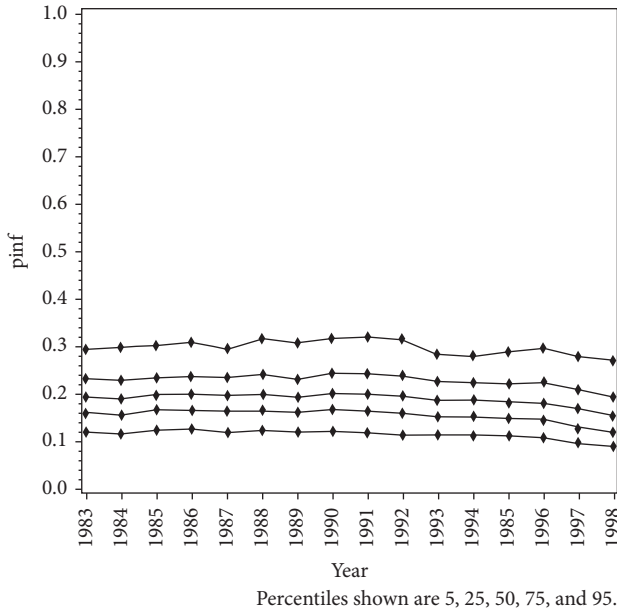


Figure 5.5 Estimated median PIN for selected trading volume deciles.

Source: panel A of fig. 3 in Easley et al. 2002, p. 2204.

dealers to tell apart informed and uninformed traders (see Chapter 8 for further discussion). Accordingly, when an anonymous and a non-anonymous trading venue coexist, informed traders should choose the anonymous one. The German Stock Exchange offers an ideal laboratory for testing this hypothesis, because its stocks are traded in both an electronic market and a floor market. Using data on trades in both markets, Grammig, Schiereck, and Theissen (2001) estimate the PIN for thirty stocks. As predicted, they find that the PIN is higher in the electronic market for almost all the stocks. Exercise 5 in this chapter is based on their study. Easley, O'Hara, and Paperman (1998) gives another application, showing that the PIN is higher for stocks with less analyst coverage, which may be the reason why the spread widens when analysts stop following a stock (as found by Ellul and Panayides 2011).

PIN measures have become so widely used that some researchers make their estimates of the PIN for different stocks publicly available on their websites (see, for instance, the website maintained by Stephen Brown at the University of Maryland: <https://terpconnect.umd.edu/stephenb/>).¹²

¹² Other applications of this methodology include Easley et al. (1996a, b); Easley, O'Hara, and Paperman (1998); Easley et al. (2001); Easley et al. (2002); and Heidle and Huang (2002). There is an ongoing debate on the extent to which PIN models match the variability of noise trade in the data and thus provides an accurate measure of informed trading; see Duarte, Hu, and Young (2020) for a discussion of alternative PIN measures and their relative merits.

5.5 Further Reading

This chapter is actually only an introduction to some empirical techniques in the field of market microstructure. For a systematic treatment, see Hasbrouck (2007).

5.6 Exercises

- 1. Price impact regressions with order flow predictability.** The goal of this exercise is to derive equation (5.15) using the Glosten and Milgrom model analyzed in Chapter 3. We use the same notations as in Chapter 3. Remember that in Chapter 3, we have shown (see equation (3.24)) that:

$$\mu_t = \mu_{t-1} + s(d_t)d_t,$$

with $s(+1) = s_t^a$ and $s(-1) = s_t^b$ given in equation (3.13) and equation (3.14). Moreover, in exercise 11 of Chapter 3, it is shown that:

$$E(d_t | \Omega_{t-1}) = \pi(2\theta_{t-1} - 1).$$

Define:

$$\lambda_t = \frac{2\pi\theta_{t-1}(1 - \theta_{t-1})(v^H - v^L)}{(1 - \pi(1 - 2\theta_{t-1}))(1 + \pi(1 - 2\theta_{t-1}))}.$$

- a.** Using the definition of s_t^a , show that:

$$\mu_t = \mu_{t-1} + \lambda_t(1 - E(d_t | \Omega_{t-1})).$$

- b.** Using the definition of s_t^b , show that:

$$\mu_t = \mu_{t-1} + \lambda_t(-1 - E(d_t | \Omega_{t-1})).$$

- c.** Deduce that:

$$\mu_t = \mu_{t-1} + \lambda_t(d_t - E(d_t | \Omega_{t-1})).$$

That is, in line with intuition, dealers' revision of the fair value of the asset are proportional to the innovation in the order flow.

- 2. The information effect of orders.** You have a record of transaction prices for all trades on March 26, 2001, for AGF—a French stock listed on

Euronext. There were 519 transactions (trading in Euronext is continuous from 9:00 a.m. to 5:25 p.m.). The index t denotes the t^{th} transaction. The data set contains the following time series (the data are available in two different formats Ch5_AGF_data.xls or Ch5_AGF_data.dta in the companion website of the book.): (i) transaction prices (“trprice”), (ii) the best ask price posted just before the transaction (“ask”), (iii) the best bid price posted just before the transaction (“bid”), (iv) the size of transactions (“tradesize”), and (v) the direction of each trade (“tradedir”).

- a. Estimate equation (5.22) and interpret the findings.
 - b. Estimate a VAR model such as equation (5.24) (accounting only for trade direction, not size, for comparability with the findings in a). **Note:** Be careful with the timing convention: m_t is the midquote before the t^{th} transaction in the model.
 - c. Describe the average dynamics of the midquote for AGF after a sell order and infer the estimate of $\alpha_{10}(1)$.
 - d. How do you explain the difference between the measure of adverse selection obtained in (a) and in (b)?
3. **Empirical specification of price impact regressions.** Suppose that the midprice at time t , m_t , is determined by the best estimate of the stock’s fundamental value based on the public information available, namely $\mu_{t-1} + \varepsilon_t$, minus a term reflecting the net inventory position z_t of market makers at time t :

$$m_t = \mu_{t-1} + \varepsilon_t - \beta z_t,$$

where the inventory at time t is related to the order flow at time $t - 1$ by the identity:

$$z_t = z_{t-1} - q_{t-1}.$$

When market makers receive an order q_t at time t , they update their estimate of the fundamental value $\mu_{t-1} + \varepsilon_t$ to also reflect the innovation in order flow, $q_t - E[q_t | \Omega_{t-1}]$:

$$\mu_t = \mu_{t-1} + \lambda \{q_t - E[q_t | \Omega_{t-1}]\} + \varepsilon_t.$$

Finally, the order flow is generated by the following AR(2) process:

$$q_t = \varphi_1 q_{t-1} + \varphi_2 q_{t-2} + \eta_t,$$

where η_t is a zero-mean error term uncorrelated with all information at time t (i.e., all variables in the information set Ω_{t-1}).

- a. Define the unexpected component of the order flow, $q_t - E[q_t | \Omega_{t-1}]$ as a function of the current and past values of the order flow.
 - b. Determine the change in the best estimate of the fundamental value, $\Delta\mu_t$.
 - c. Determine the change in the midprice, Δm_t , as a function only of past values of the order flow.
 - d. If you estimate the equation for Δm_t obtained at point (c) jointly with the order flow process assumed in equation $q_t = \varphi_1 q_{t-1} + \varphi_2 q_{t-2} + \eta_t$, can you identify the parameters of the model, namely, β , λ , φ_1 , and φ_2 ? If so, explain how you would infer their values, denoting the coefficients of first three lags of the order flow in the equation obtained at point (c) by b_0 , b_1 , and b_2 .
 - e. Do we actually have over-identifying restrictions that can be tested? Specifically, does the above model imply a testable restriction on b_1/b_2 ?
4. **Price impact estimation.** This exercise is based on data collected by Richard Lyons, who used these data to estimate a model of trading with inventory effects (see Lyons 1995). The data are stored in the Excel file Ch5_ex4_data.xls and in the Stata data file Ch5_ex4_data.dta available in the companion website for the book. The data set covers one week of trading (August 3–7, 1992) of one dealer, in the Deutsche Mark/Dollar market, at a major New York investment bank. The data set has two components:
1. A time series of all prices and quantities for all **direct, incoming** transactions in which the dealer was involved. “**Direct**” means that the transactions result from direct bilateral negotiations between the dealer and another dealer (in contrast to brokered transactions). “**Incoming**” means that the dealer does not initiate the transaction (that is, he acts as liquidity supplier).
 2. A time series of the dealer’s inventory. This inventory is observed at the time of each incoming transaction.
- There are 843 observations for the week. Quotes in the Deutsche Mark/-dollar market were firm up to \$10 million. The minimal price increment in this market is .0001DM and is referred to as “one pip.” Lyons (1995) reports that the median spread quoted by the dealer is three pips. The dataset includes four time series:
1. **price:** transaction prices (p_t).
 2. **inventories:** the dealer’s inventory (I_t) in million U.S. dollars. I_t is the dealer’s inventory just before the t^{th} transaction.
 3. **tradesize:** the trade sizes (Q_t) in million U.S. dollars. Trades are signed according to the position taken by the liquidity demander. Purchases (i.e., dealer’s sales) are positive, and sales (i.e., dealer’s purchases) are negative.

4. **signedtr**: an indicator variable (d_t) that is equal to +1 for buy orders (sales by the dealer) and -1 for sell orders (purchases for the dealer). Overnight observations are suppressed since microstructure effects explain only intraday price dynamics, not overnight changes.
 - a. Plot the time series of dealer's inventory at the time of each incoming order in the sample. How do you interpret the graph? Is there evidence of mean reversion in inventories?
 - b. Use the data to estimate equation (5.23). How do you interpret the findings?
 - c. Propose an estimate of the cost of trading \$50 million with the dealer.
5. **PIN estimation**. This exercise uses data from the German Stock Exchange (a subset of the data set used in Grammig, Schiereck, and Theissen 2001).¹³ The data set gives, for one stock (BVM) traded on the German Stock Exchange, the number of buy orders and sell orders from June 2 to July 31, 1997 (forty-two days). As usual, the orders are signed according to the position taken by liquidity demanders (i.e., an order is signed positively when the trade initiator is a buyer and conversely). The stock trades on two markets that operate in parallel: a floor market and an electronic trading system. The main difference between the two systems is relative anonymity: a floor market is less anonymous as traders on the floor negotiate prices one-on-one. The data are stored in the Excel file Ch5_ex5_data.xls and the Stata file Ch5_ex5_data.dta available on the companion website for the book. This dataset contains four time series:
 1. **buy_f**: The number of buy orders executed in the floor market on each day t , $t \in \{1, \dots, 42\}$.
 2. **sell_f**: The number of sell orders executed in the floor market on each day t , $t \in \{1, \dots, 42\}$.
 3. **buy_e**: The number of buy orders executed in the electronic market on each day t , $t \in \{1, \dots, 42\}$.
 4. **sell_e**: The number of sell orders executed in the electronic market on each day t , $t \in \{1, \dots, 42\}$.
 - a. Consider the tree describing the order arrival process in Figure 5.3. We modify it to account for the possibility that traders can choose to trade in either the floor market or the electronic market. Specifically, we suppose that informed investors trade at rate μ_F in the floor market and at rate μ_E in the electronic market, while uninformed investors buy and sell at rates ε_{bj} and ε_{sj} in market $j \in \{E, F\}$. On these assumptions, what is the likelihood that this trade is informed, conditional on a trade taking place in the floor market? Conditional on a trade taking place in the electronic market? Call these likelihoods PIN_F and PIN_E .

¹³ We thank Erik Theissen for providing the data used here.

- b. Using the series of buy and sell orders executed in each system, propose and implement a methodology to estimate PIN_F and PIN_E .
(Note: Estimation by maximum likelihood may not converge if the initial values for the parameters are not well chosen. For this you must calibrate the initial values of the parameters to estimate, so that at least the average number of buy and sell orders per day on each market implied by the Poisson distributions match the actual averages in the data.)
- c. Are informed traders more likely to trade in the anonymous market? Is adverse selection greater in the anonymous market?
- 6. **Decomposition of the bid-ask spread.** Using the data for exercise 1 (Ch5_AGF_data.xls or the Stata file Ch5_AGF_data.dta available in the companion website for the book) and following Glosten and Harris (1988), regress Δp_t on d_t , q_t , Δd_t , and Δq_t .
 - a. For a transaction of very small size, what is the bid-ask spread and what proportion is attributable to adverse selection versus order-processing cost?
 - b. Use an F-test to check whether the data support the restriction (imposed by Glosten and Harris 1988) that the coefficients of Δq_t and d_t must both be zero.
 - c. Use an F-test to check whether the trade size effects (as measured by the coefficients of q_t and Δq_t) are jointly significant over and above the impact of the buy/sell indicators d_t and Δd_t .
 - d. Relate your answers to the spread measures calculated in the exercises of Chapter 2.
- 7. **Bid-ask spread and PIN.** Consider the order arrival process described in Figure 5.3. Using equation (3.8) in Chapter 3, derive the ask and bid prices for the first transaction of the day.
- 8. **Method-of-moments estimator for PIN.** For the PIN model as described in Section 5.4:
 - a. Derive the first and second moments of the daily buy and sell transactions, that is the expectations $E[B]$ and $E[S]$, the variances $\text{var}[B]$ and $\text{var}[S]$ and the covariance $\text{cov}[B, S]$, as a function of the five underlying parameters of the model (α , θ , ϵ_b , ϵ_s , ϵ_ϵ).
 - b. Use these five equations to express PIN as well as the underlying parameters of the model in terms of these moments.